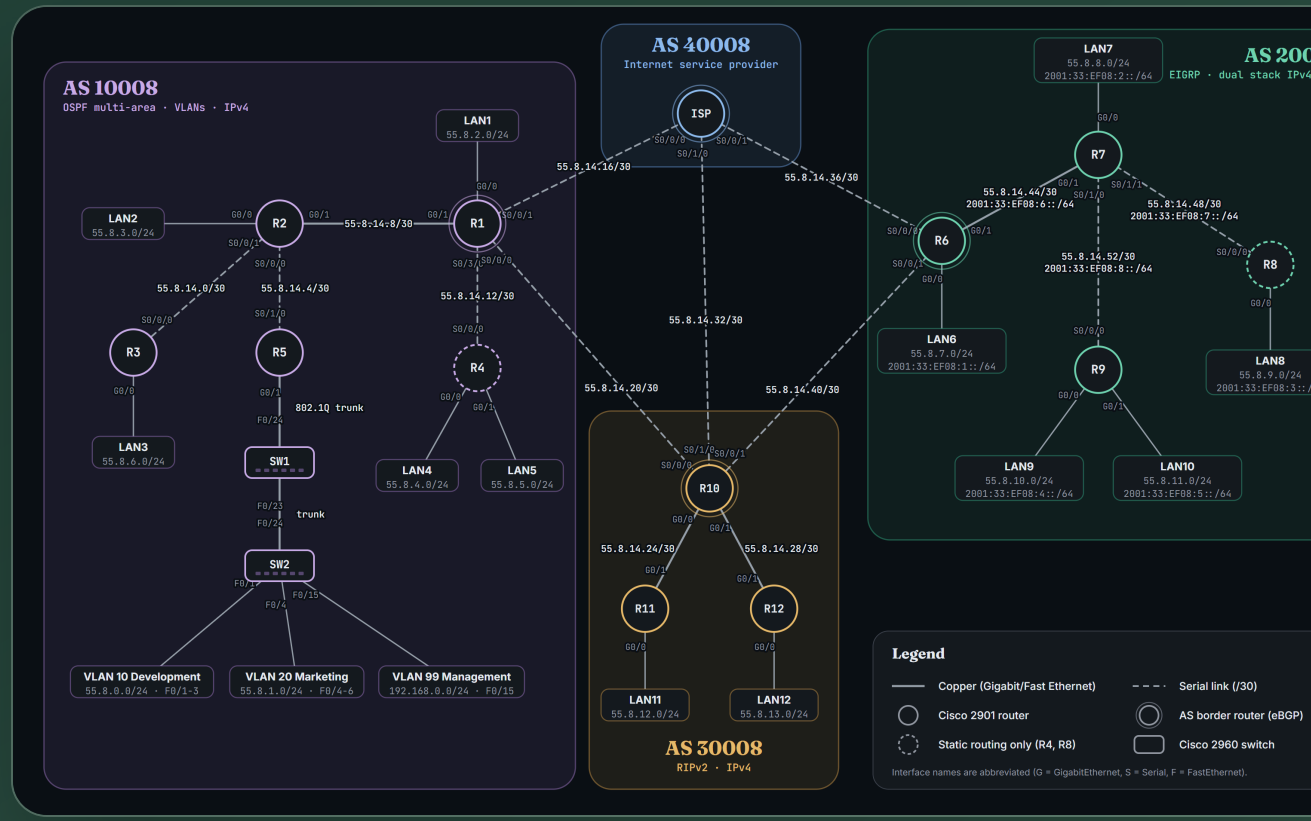
FINAL PROJECT · INTERNET ARCHITECTURE

Autonomous Systems

An Internet-like network in Cisco Packet Tracer: four autonomous systems, five routing techniques, two IP versions.

OSPF multi-area · EIGRP IPv4/IPv6 · RIPv2 · eBGP · 802.1Q VLANs · static routing



Lazar Davidović · Cisco Packet Tracer 8.1.1

THE BRIEF

Connect, address and route a small Internet

4
autonomous systems
3 customers + 1 ISP

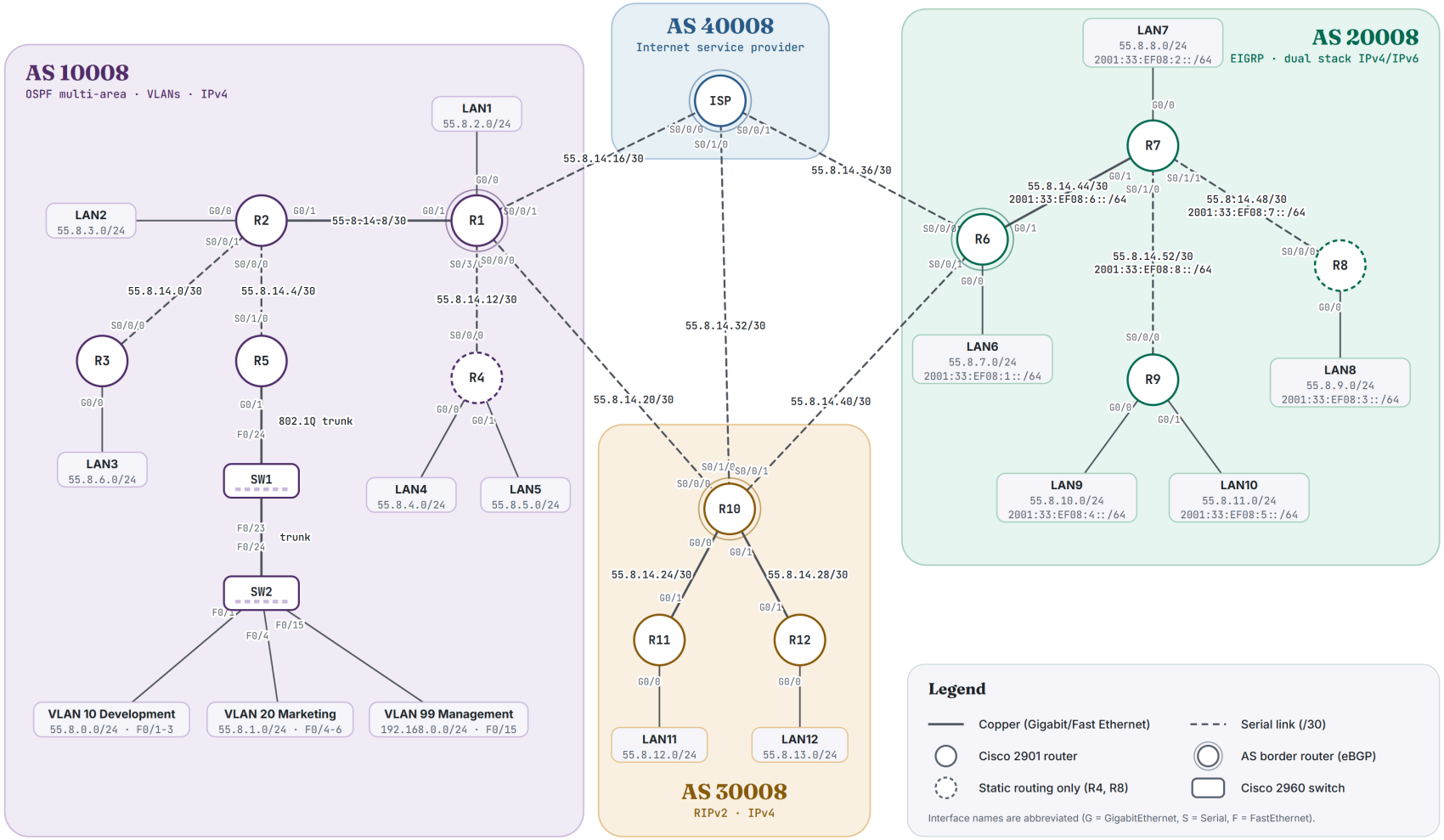
13
Cisco 2901 routers
+ 2 Cisco 2960 switches

31
links: 10 serial /30
and 21 copper

14
user segments: 12 LANs
and 2 routed VLANs

AS	Interior routing	Special requirement
10008	multi-area OSPF	three VLANs behind R5 · R4 on static routes
20008	EIGRP for IPv4 and IPv6	the only dual-stack AS · R8 on static routes
30008	RIPv2	–
40008	– (ISP)	eBGP to every customer; ISP link primary, others backup

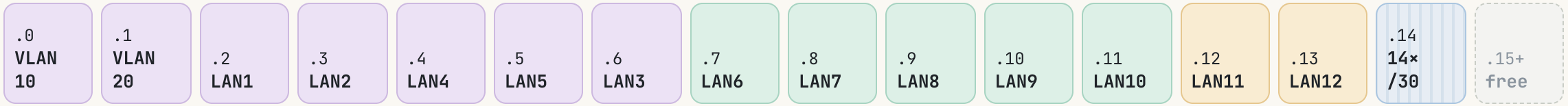
Implemented network



TASK 1 · ADDRESSING

One /16, sized for growth

55.8.*x*.0/24 — third octet



- LAN: 126 PCs + gateway = 127 > 126 usable in a /25 → **/24**
- VLAN 10/20: 200 PCs + gateway → **/24** · links → **/30**
- LAN4 + LAN5 placed at .4 and .5 → one aligned **55.8.4.0/23** for the static route and BGP

IPv6 in AS 20008

2001:33:EF08::/48 → a **/64** per segment, subnet ID = sequence number.

```
LAN6 … LAN10   2001:33:EF08:1-5::/64
R6–R7          2001:33:EF08:6::/64
R7–R8          2001:33:EF08:7::/64
R7–R9          2001:33:EF08:8::/64
link-local     FE80::n  (n = router)
```

Management VLAN: private 192.168.0.0/24

Router-on-a-stick behind R5

VLAN	Purpose	Subnet	Ports
10	Development	`55.8.0.0/24`	F0/1–3
20	Marketing	`55.8.1.0/24`	F0/4–6
99	Management	`192.168.0.0/24`	SW2 F0/15

- 802.1Q trunks R5 – SW1 – SW2
- Routing between VLAN 10 and 20 on R5 subinterfaces
- **VLAN 99 is never routed**: no subinterface on R5, private range, not advertised

```
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 55.8.0.1 255.255.255.0
!
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 55.8.1.1 255.255.255.0
! SW1 / SW2
interface FastEthernet0/1
 switchport mode access
 switchport access vlan 10
interface FastEthernet0/24
 switchport mode trunk
```

Two stub routers outside the IGP

R4 · AS 10008 · IPv4

Default route to R1; R1 points one summary at both LANs and redistributes it into OSPF.

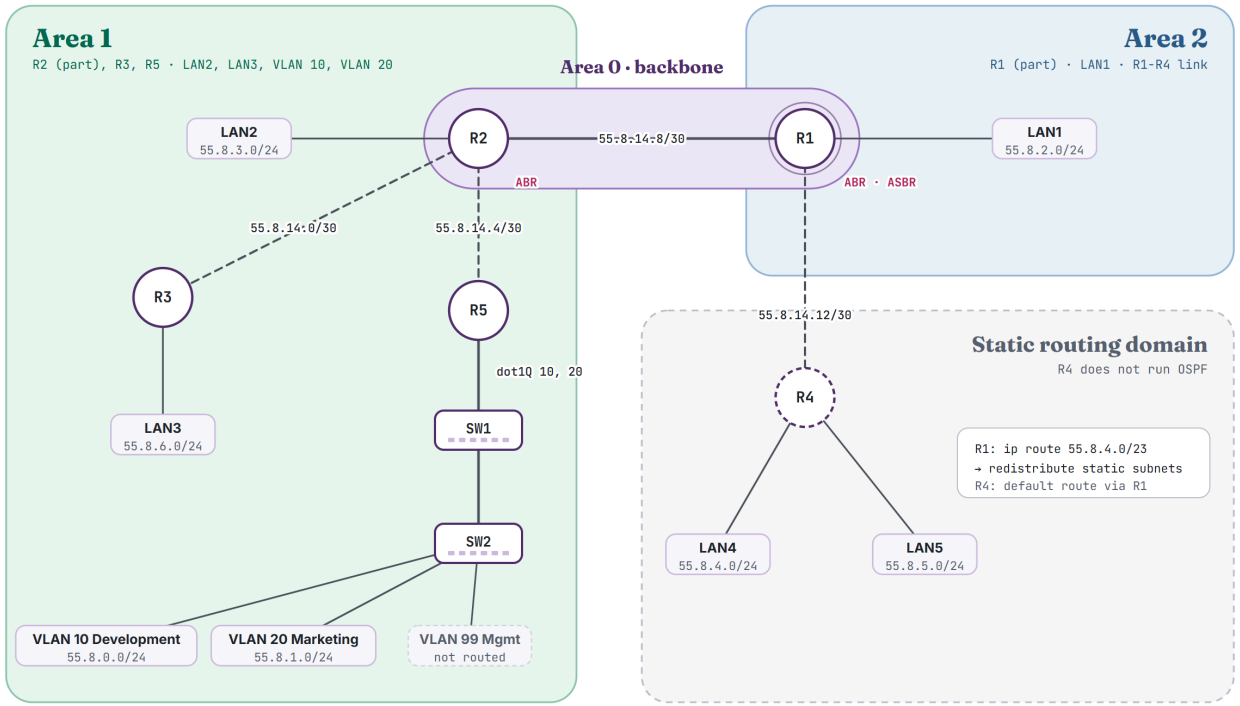
```
! R4
ip route 0.0.0.0 0.0.0.0 55.8.14.13
! R1
ip route 55.8.4.0 255.255.254.0 55.8.14.14
router ospf 8
 redistribute static subnets
```

R8 · AS 20008 · IPv4 + IPv6

The same pattern for both address families; R7 redistributes LAN8 into EIGRP.

```
! R8
ip route 0.0.0.0 0.0.0.0 55.8.14.49
ipv6 route ::/0 2001:33:EF08:7::1
! R7
ip route 55.8.9.0 255.255.255.0 55.8.14.50
ipv6 route 2001:33:EF08:3::/64 2001:33:EF08:7::2
```

Multi-area OSPF



- **Area 0**: the R1 – R2 link
- **Area 1**: R3, R5, LAN2, LAN3, both VLANs
- **Area 2**: LAN1 and the link to R4
- **R1, R2** are ABRs; **R1** is also the ASBR
- LAN4 + LAN5 enter as `O E2` `55.8.4.0/23`
- Process 8, router IDs 1.1.1.1 … 5.5.5.5

Dual-stack EIGRP

IPv4

Classic `network` statements with wildcards on R6, R7, R9. Links to the ISP and R10 stay outside.

IPv6

`ipv6 unicast-routing`, a 32-bit `eigrp router-id`, `no shutdown`, then `ipv6 eigrp 20008` per interface.

LAN 8

R7 redistributes its static routes in both processes → `D EX` routes on R6 and R9.

```
router eigrp 20008
 eigrp router-id 7.7.7.7
 redistribute static
 network 55.8.8.0 0.0.0.255
 network 55.8.14.44 0.0.0.3
 network 55.8.14.48 0.0.0.3
 network 55.8.14.52 0.0.0.3
```

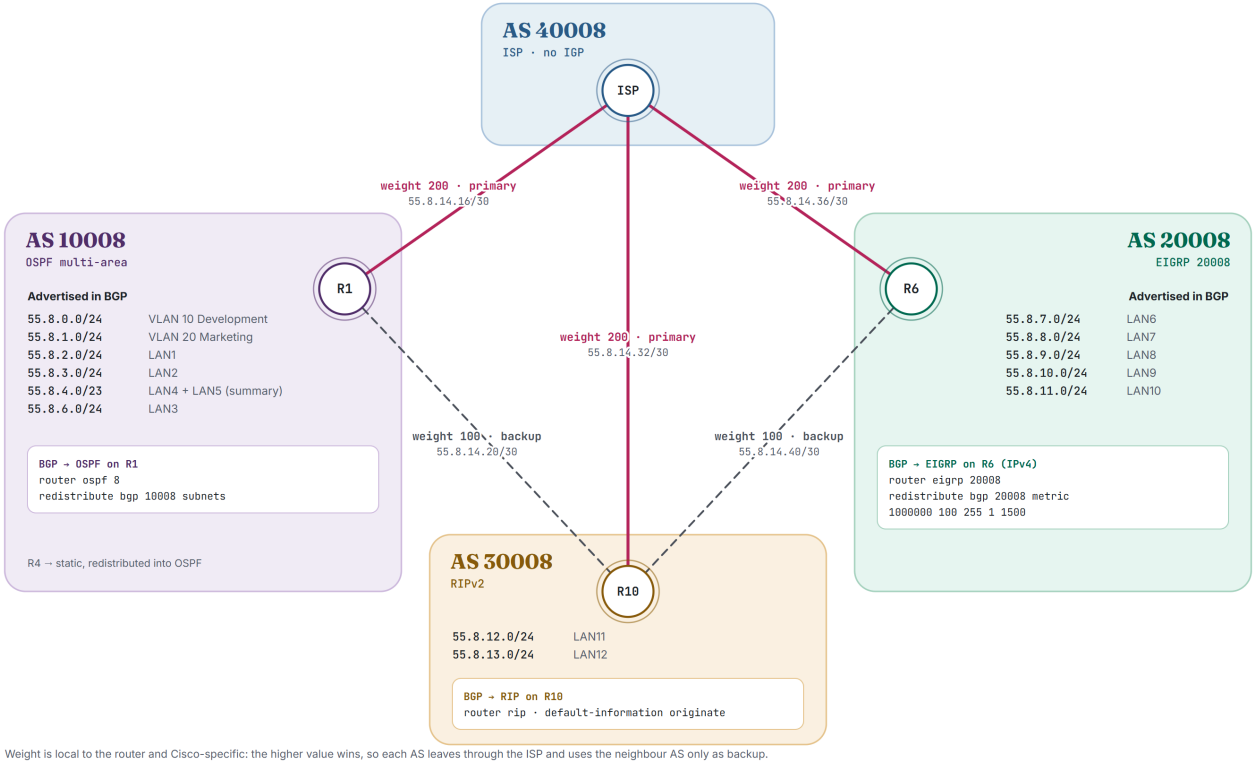
```
ipv6 router eigrp 20008
 eigrp router-id 7.7.7.7
 no shutdown
 redistribute static
!
interface GigabitEthernet0/1
 ipv6 address 2001:33:EF08:6::2/64
 ipv6 eigrp 20008
```

RIPv2: one classful statement

```
! R10, R11, R12
router rip
 version 2
 network 55.0.0.0
```

- RIP networks are classful; every address in the AS lives in **55.0.0.0/8**, so one line enables RIP everywhere
- Auto-summary has no effect for the same reason
- R10 learns LAN11 and LAN12 and later announces them in BGP
- **Improvement:** the statement also covers R10's three inter-AS links, so `passive-interface` belongs there

ISP first, neighbour AS as backup



- One border router per AS → plain eBGP, no iBGP
- **weight 200** on the ISP session, **100** on customer sessions: higher wins, local to the router
- Only LAN and VLAN prefixes are announced, never /30 links
- Packet Tracer rejects `weight`, so those lines are documented, as the brief allows

TASK 10 · REDISTRIBUTION

Bringing BGP routes inside each AS

Border router	Into	Command	Why
R1	OSPF 8	`redistribute bgp 10008 subnets`	O E2 routes; `subnets` keeps classless prefixes
R6	EIGRP 20008	`redistribute bgp 20008 metric 1000000 100 255 1 1500`	EIGRP needs a seed metric for external routes
R10	RIPv2	`default-information originate`	R11/R12 only need a default towards R10

bandwidth	delay	reliability	load	MTU
1,000,000 kbit/s	**100 × 10 µs**	**255 / 255**	**1 / 255**	**1500 B**

The five components of the EIGRP seed metric used on R6.

VERIFICATION

Configuration checked as code

- Every running-config exported from the `.pkt` into `configs/`
- `scripts/check_configs.py` checks addressing, VLANs, IGP coverage, static next hops, OSPF areas and the BGP design
- Standard library only; runs in **GitHub Actions** on every push
- In the simulator: `show ip ospf neighbor`, `show ipv6 eigrp neighbors`, `show ip route`, ping and traceroute between LANs

```
$ python scripts/check_configs.py
Devices: 15  links: 31  hosts: 15  BGP routers: 4
Checks passed: 727
WARNING  host LAN1 uses mask /25, router uses /24
WARNING  admin PC has no IP address
WARNING  R10: RIP updates on inter-AS links
OK
```

STATUS

What the saved simulation contains

Implemented in the .pkt

- Addressing and interfaces (tasks 1–2)
- VLANs and inter-VLAN routing (3)
- Static routing on R4 and R8 (4, 6)
- OSPF, EIGRP for IPv4 and IPv6, RIPv2 (5, 7, 8)

Designed and documented

- eBGP and redistribution (9–10): `configs/bgp/`
- Remote switch management: admin PC address and VTY passwords
- Clean-ups: LAN1 PC mask, `passive-interface` on inter-AS and stub links

REPOSITORY

Thank you

Topology, configurations, report and checker:

`github.com/LazarDavidovic98/internet-architecture-project`

`src/ai-project.pkt` `configs/` `docs/Report.pdf` `scripts/check_configs.py`